

Project	GameFi Economy — ERC-1155 In-Game Item Economy with Crafting, Governance & Rental Vault
Stack	Solidity ^0.8.24 / ^0.8.25 · Foundry · Chainlink VRF v2.5 · Chainlink Price Feeds · OpenZeppelin v5 · The Graph
Network	Arbitrum Sepolia (primary testnet) · Base Sepolia (L2 compat) · Foundry Anvil (local)
Authors	Roman (Smart Contracts) · Anatoly (Architecture / Frontend / Subgraph)
Repository	https://github.com/Fipaان/bt2-final.git

Executive Overview

GameFi Economy is a fully on-chain gaming protocol built around a custom ERC-1155 implementation (Equestria1155) that unifies fungible element tokens (HONESTY, KINDNESS, LAUGHTER, GENEROSITY, LOYALTY, MAGIC — IDs 1–6) and semi-fungible pony NFTs (PINKIE_PIE id 1001, STARLIGHT_GLIMMER id 1002) in a single contract. Players earn ponies through Chainlink VRF v2.5-powered crafting (burning element tokens → minting pony NFTs with a 10% double-drop chance), trade ERC-20 game tokens on a constant-product AMM, and stake rare ERC-1155 NFTs in an ERC-4626 rental vault for yield boosts. All protocol economics — vesting, treasury, upgrade authority — are controlled by a GovernanceToken (GOV) DAO with a TimelockController.

Core Standard	Description
ERC-1155	Single Equestria1155 contract manages all fungible element tokens (HONESTY, KINDNESS, LAUGHTER, GENEROSITY, LOYALTY, MAGIC; IDs 1–6) and semi-fungible ponies (PINKIE_PIE id 1001, STARLIGHT_GLIMMER id 1002)
Randomness	Chainlink VRF v2.5 (VRFConsumerBaseV2Plus) — verifiable loot drops inside Equestria1155.craft(); 10% chance of double drop; Standalone VRFConsumer for general random requests
Crafting	Equestria1155.craft() burns input element tokens per Recipe struct, requests VRF, then fulfillRandomWords mints pony NFT (1 or 2 on lucky roll). Two recipes defined: PINKIE_PIE and STARLIGHT_GLIMMER
AMM	Constant-product AMM (0.3% fee) for two ERC-20 GameToken pairs (e.g. HAR/SPK). Uniswap v2-style addLiquidity/removeLiquidity/swap with Yul assembly getAmountOutYul optimisation
Rental Vault	ERC-4626 NFTRentalVault accepts GovernanceToken as asset; ERC-1155 NFT stakers earn a configurable boost (default 120%). Owner (Timelock) injects yield via injectYield()
DAO Governance	GovernanceToken (ERC-20Votes, 1M total supply) + MyGovernor (OZ Governor extensions) + TimelockController. Token split: 40% team vesting, 30% treasury, 20% airdrop, 10% liquidity
Factory	GameTokenFactory deploys ERC-20 GameTokens via CREATE (unpredictable) or CREATE2 (deterministic). UUPS upgradeable GameTokenV1/V2 proxy pattern also demonstrated
L2 Deployment	Arbitrum Sepolia: VRF_COORDINATOR 0x5CE8D5A2..., VRF_KEY_HASH 0x1770bdc7..., ETH/USD feed 0xd30e2101... Configurable staleness threshold (10 min testnet / 1 h production)

ERC-1155 Token Design (Equestria1155)

The Equestria1155 contract manages the entire in-game token economy. Unlike the generic GameFi template, the actual implementation uses a My Little Pony-themed element/pony model with two hardcoded crafting recipes. Token IDs are small fixed integers (not range-partitioned), and all randomness flows through Chainlink VRF v2.5 during the craft() flow.

Token ID	Name / Type	Description	Tradeable on AMM
1	HONESTY (Fungible)	Element token — ingredient for crafting recipes	No (used in crafting)
2	KINDNESS (Fungible)	Element token — ingredient for crafting recipes	No
3	LAUGHTER (Fungible)	Element token — ingredient; 100 needed for PINKIE_PIE	No
4	GENEROSITY (Fungible)	Element token — ingredient for crafting recipes	No
5	LOYALTY (Fungible)	Element token — ingredient for crafting recipes	No
6	MAGIC (Fungible)	Element token — ingredient; 80 needed for STARLIGHT_GLIMMER	No
1001	PINKIE_PIE (Semi-fungible)	Recipe: 15H + 10K + 100L + 20G + 5Lo + 0M. Stackable, minted by VRF	Via Rental Vault
1002	STARLIGHT_GLIMMER (Semi-fungible)	Recipe: 5H + 3K + 0L + 0G + 15Lo + 80M. Stackable, minted by VRF	Via Rental Vault

Crafting Flow

1. Player calls craft(ponyId) — Equestria1155 validates the recipe exists (sum > 0 guard), checks element balances, burns elements from caller, and calls requestRandomWords on the VRF coordinator.
2. Chainlink VRF calls fulfillRandomWords() — the internal callback looks up pendingCraftUser and pendingCraftPonyId, deletes them, then mints 1 or 2 pony NFTs (10% double-drop if randomWord % 100 < 10).

Core Smart Contracts

Contract	Standard / Pattern	Responsibility
Equestria1155	ERC-1155 + VRFConsumerBaseV2Pluses	All element/pony token IDs; craft() burns elements → requests VRF → fulfillRandomWords mints pony
AMM	Constant-product AMM (custom)	Swap between two ERC-20 GameToken addresses; 0.3% fee; getAmountOutYulYul optimisation; LP token minting/burning
GameToken	ERC-20 + Ownable	Simple mintable game token; owner-controlled; used as AMM pair tokens (HAR, SPK) and also as GovernanceToken base

LPToken	ERC-20	AMM LP share token; deployed by AMM constructor; only AMM can mint/burn
NFTRentalVault	ERC-4626 + Ownable + ReentrancyGuard	Accepts GovernanceToken as asset; ERC-1155 stakers get share boost; owner injects yield; IERC1155Receiver
GovernanceToken	ERC-20Votes + ERC-20Permit	1M GOV; immutable distribution at deploy: 40% vesting, 30% treasury, 20% airdrop, 10% liquidity
MyGovernor	OZ Governor + 5 extensions	Quorum 4%; proposalThreshold = 1% of supply; votingDelay=5 blocks, votingPeriod=10 blocks (testnet); Timelock-controlled execution
Treasury	Custom + Ownable (Timelock)	ETH + ERC-20 treasury; all withdrawals require Timelock (DAO vote); receive() payable
TokenVesting	Custom linear vesting	365-day linear vesting schedule; release() callable by anyone; NothingToRelease guard; SafeERC20
Box	Custom + Ownable (Timelock)	Simple DAO-controlled storage contract; demonstrates proposal execution pattern
GameTokenFactory	Factory + CREATE2	createToken() (CREATE) and createToken2() (CREATE2) deploy GameToken; predictAddress() computes future address
GameTokenV1 / V2	ERC-20Upgradeable + UUPS	Proxy upgrade pattern demo; _disableInitializers() in constructor; V2 adds burn(); _authorizeUpgrade onlyOwner
VRFConsumer	VRFConsumerBaseV2Plu s	Standalone general-purpose VRF consumer; stores requestId→sender; emits RandomRequested/RandomFulfilled
PriceFeedConsumer	Chainlink AggregatorV3Interface	getLatestPrice() with staleness check, invalid round check (answeredInRound), and negative price guard

External Integrations

Integration	Usage	Safety Measure
Chainlink VRF v2.5	Loot drops in Equestria1155.craft(); standalone VRFConsumer; VRFV2PlusClient.RandomWordsRequest	Subscription-funded; CALLBACK_GAS_LIMIT 200k (craft) / 100k (standalone); fulfillRandomWords internal override; only VRF coordinator can call
Chainlink Price Feeds	PriceFeedConsumer.getLatestPrice() — AggregatorV3Interface; ETH/USD feed on Arbitrum Sepolia	Staleness check (configurable threshold, default 10 min testnet); InvalidRound if startedAt==0 or answeredInRound < roundId; InvalidPrice if price < 1
OpenZeppelin v5	Governor, TimelockController, ERC1155, ERC4626, ERC20Votes, ERC20Permit, UUPSUpgradeable,	Pinned versions in foundry.toml and foundry.lock; via_ir=true; optimizer_runs=200

	AccessControl, ReentrancyGuard, SafeERC20, Math	
The Graph	Subgraph indexes ERC-1155 Transfer, Craft, Swap, Deposit, VRF events; frontend uses Apollo Client	Hosted on The Graph Studio; schema.graphql + AssemblyScript mappings

Deployment Strategy

Deployment Order (Deploy.s.sol)

Step	Action	Notes
1	Deploy GovernanceToken(deployer×4)	Mints 40/30/20/10% to team, treasury, airdrop, liquidity addresses (all deployer on testnet)
2	Deploy TokenVesting(token, deployer, now)	365-day linear vesting for team allocation; start = block.timestamp
3	Deploy TimelockController(10s, [], [0x0], deployer)	10s delay for testnet (use 2d+ for mainnet); open executor (address(0)); deployer is admin
4	Deploy MyGovernor(token, timelock)	Grant PROPOSER_ROLE to governor; EXECUTOR_ROLE to address(0) (anyone can execute passed proposals)
5	Deploy Treasury(timelock) + Box(timelock)	Both contracts owned by Timelock — all state changes require DAO vote
6	Deploy GameToken×2 (HAR, SPK) + AMM	HAR (Harmony) and SPK (Sparkle) as pair tokens; AMM deploys its own LPToken
7	Deploy Equestria1155(uri, VRF_COORD, keyHash, subId)	Uses Arbitrum Sepolia Chainlink VRF coordinator; PINKIE_PIE and STARLIGHT_GLIMMER recipes pre-configured
8	Deploy GameTokenFactory + GameTokenV1/V2 + ERC1967Proxy	UUPS proxy pattern demo; proxy initialised with GameTokenV1; upgradeability restricted to proxy owner
9	Deploy NFTRentalVault(token, nft, PINKIE_PIE, timelock)	GovernanceToken as vault asset; PINKIE_PIE (id 1001) as boosted NFT; owner = Timelock
10	Deploy PriceFeedConsumer(PRICE_FEED, 10min) + VRFConsumer	ETH/USD Arbitrum Sepolia feed 0xd30e2101...; VRF standalone for general randomness demos

Network Targets

Environment	Network	Purpose
Local	Foundry Anvil	Unit / fuzz / invariant / fork tests (80+ test cases across 11 test files)
Testnet (primary)	Arbitrum Sepolia	Integration, Chainlink VRF v2.5, price feeds, subgraph QA; VRF_SUB_ID from env

Testnet (compat)	Base Sepolia	L2 compatibility check, gas profiling
CI/CD	GitHub Actions (.github/workflows/test.yml)	forge build → forge test → Slither (--exclude-dependencies --filter-paths lib/) → gas report

Upgrade & Proxy Strategy (UUPS)

The project demonstrates UUPS upgradeability through the GameTokenV1 / GameTokenV2 pair. A single ERC1967Proxy is deployed pointing at GameTokenV1; the owner calls upgradeTo(implV2) to activate V2 features (burn function). Key safety measures:

- `_disableInitializers()` in constructor prevents implementation contract exploitation.
- `_authorizeUpgrade(address)` restricted to owner — must be Timelock in production.
- `initialize()` with initializer modifier replaces constructor; prevents re-initialisation.
- GameTokenV2 adds a `burn()` function, demonstrating additive upgrades without storage collision.

Security Considerations

Risk	Mitigation in Codebase
ERC-1155 batch reentrancy	ReentrancyGuard on Equestria1155.craft() and NFTRentalVault.stakeNFT/unstakeNFT; CEI pattern throughout; Slither reentrancy false-positives annotated
Oracle manipulation / staleness	PriceFeedConsumer checks <code>block.timestamp - updatedAt > stalenessThreshold</code> , <code>answeredInRound < roundId</code> , <code>startedAt == 0</code> , and <code>price < 1</code>
VRF front-running / access	<code>fulfillRandomWords</code> is internal override, only callable by VRF Coordinator; <code>requestId</code> validated via <code>pendingCraftUser</code> mapping; delete on fulfillment
Integer overflow	Solidity ^0.8.x native checks throughout; unchecked <code>{}</code> only in audited tight loops (e.g. <code>_balances</code> subtraction after bounds check)
Crafting recipe exploit	Recipes are hardcoded at deploy in constructor (PINKIE_PIE, STARLIGHT_GLIMMER); <code>total = 0</code> guard reverts <code>InvalidRecipe()</code> ; elements burned before VRF call
AMM pool drain	<code>getAmountOut</code> reverts if <code>amountOut >= reserveOut</code> (strict inequality prevents full drain); slippage parameter on all write functions
UUPS upgrade authority	GameTokenV1/V2 <code>_authorizeUpgrade</code> restricted to owner; in production owner should be Timelock; initializer modifier prevents re-init
DAO centralisation	Treasury, Box, NFTRentalVault all owned by TimelockController; no privileged EOA bypass post-deploy; <code>proposalThreshold = 1%</code> of supply
Burn-before-VRF atomicity	<code>craft()</code> burns all element tokens, then calls <code>requestRandomWords</code> ; if VRF call fails the burn is rolled back (Solidity revert semantics)

Indexing & Frontend

Layer	Technology	Key Responsibilities
Subgraph	The Graph (schema.graphql + mappings)	Index Transfer, CraftRequested, CraftFulfilled, Swap, NFTStaked, NFTUnstaked, YieldInjected, VRF events → GraphQL
Frontend	Next.js 14 + wagmi v2 + Apollo Client	Pages: /amm, /craft, /dao, /inventory, /vault, /vesting. Wallet connect via wagmi; read balances; write txns; proposals UI
Write Txns	wagmi useWriteContract	craft(), swap(), addLiquidity(), removeLiquidity(), stakeNFT(), unstakeNFT(), deposit(), withdraw(), propose(), castVote()
Read Data	Apollo Client → The Graph Studio	Pull inventory, proposals, AMM prices, vault state, vesting schedule from subgraph
ABIs / Addresses	dapp/src/lib/contracts.ts (~216KB)	Auto-generated ABI and deployed address map for all contracts; imported by all pages

Security Audit

1. Executive Summary

This report presents findings of a security audit conducted on the GameFi Economy protocol — an on-chain gaming system consisting of a custom ERC-1155 crafting contract (Equestria1155), a constant-product AMM, an ERC-4626 NFT rental vault, a full DAO governance stack, a token vesting contract, and a UUPS proxy upgrade demo.

The audit covers manual code review, automated static analysis (Slither), and fuzz/invariant testing via Foundry. The objective is to identify vulnerabilities that could result in loss of funds, unauthorised minting, governance manipulation, or oracle exploitation.

Total findings: 0 Critical, 1 High, 2 Medium, 1 Low, 1 Informational

2. Scope

Contract	Standard	Proxy
Equestria1155	ERC-1155 + VRFConsumerBaseV2Plus	No
AMM + LPToken	Constant-product AMM	No
NFTRentalVault	ERC-4626 + IERC1155Receiver	No
GovernanceToken	ERC-20Votes + ERC-20Permit	No
MyGovernor	OZ Governor (5 extensions)	No
Treasury	Custom (ETH + ERC-20 withdrawals)	No
TokenVesting	Linear vesting + SafeERC20	No
Box	Custom (Timelock-gated storage)	No
GameTokenFactory	CREATE + CREATE2 factory	No
GameTokenV1 / V2	ERC-20Upgradeable + UUPS	ERC-1967 (UUPS)

PriceFeedConsumer	Chainlink AggregatorV3Interface	No
VRFConsumer	VRFConsumerBaseV2Plus	No

3. Methodology

Phase 1 — Automated Analysis

Slither static analyzer runs against all contracts (slither . --exclude-dependencies --filter-paths "lib/"). Slither false-positives in the codebase are annotated with // slither-disable-next-line comments. Mythril can be applied for symbolic execution on critical paths (craft(), AMM.swap(), fulfillRandomWords).

Phase 2 — Manual Review

Line-by-line review of all 13 contracts (src/) with focus on: access control, reentrancy, arithmetic, oracle integration, VRF callback security, ERC-1155 batch operations, ERC-4626 share accounting, and UUPS upgrade authority.

Phase 3 — Dynamic Testing

Foundry fuzz and invariant tests (80+ test cases across 11 test files): AMMTest invariant (constant-product $k \geq$ initial), Equestria1155 balance invariants, NFTRentalVault share/asset ratio, VRFConsumer callback access control, GovernanceToken full DAO flow (FullFlow.t.sol).

Phase 4 — Fork Testing

Fork tests against Arbitrum Sepolia (rpc_endpoints.sepolia in foundry.toml) to validate live Chainlink VRF v2.5 and price feed integration.

3.2 Severity Classification

Severity	Definition
Critical	Direct loss of funds, unauthorized minting, or complete protocol compromise
High	Significant risk; likely exploitable under specific conditions
Medium	Limited impact or requires unusual preconditions
Low	Best-practice violation; low exploit probability
Informational	Code quality, gas optimisation, documentation improvements

4. Findings Table

4.1. Slither Findings

ID	Contract	Severity	Slither warnings	Code / Condition	Justification
F-01	src/amm/AMM.sol	MEDIUM	divide-before-multiply	(amountADesired * reserveB) / reserveA	Intentional Uniswap V2-style truncation behavior is acceptable here.
F-02	src/amm/AMM.sol	MEDIUM	divide-before-multiply	(amountBDesired * reserveA) / reserveB	Intentional Uniswap V2-style truncation behavior is acceptable here.
F-03	src/dao/TokenVesting.sol	INFO	timestamp	block.timestamp < start	Timestamp comparison is required for vesting schedule validation; timestamp is clamped to expected range.

F-04	src/dao/TokenVesting.sol	INFO	timestamp, incorrect-equality	amount == 0	Zero-amount release must explicitly revert; strict comparison is intentional and justified.
F-05	src/dao/Treasury.sol	INFO	low-level-calls	to.call{value: amount}("")	Custom ETH withdrawal implementation intentionally uses low-level call.
F-06	src/vault/NFTRentalVault.sol	INFO	reentrancy-events	SafeERC20.safeTransferFrom(...)	Event emission order is intentional; emitting before external transfer would be incorrect.
F-07	src/chainlink/PriceFeedConsumer.sol	INFO	timestamp	block.timestamp - updatedAt_ > stalenessThreshold	Timestamp comparison is required to validate Chainlink price freshness.
F-08	src/chainlink/VRFConsumer.sol	INFO	reentrancy-benign	requestRandomWords(...)	False positive; external VRF request must happen before related state/event flow.
F-09	src/amm/AMM.sol	INFO	assembly	assembly ("memory-safe")	Assembly is intentionally used for performance comparison purposes.
F-10	src/amm/Equestria1155.sol	INFO	missing-zero-check	setGovernor(...)	False positive; nonZeroAddress modifier already validates the address.
F-11	src/amm/Equestria1155.sol	INFO	reentrancy-benign	requestRandomWords(...)	False positive; request ordering around external call is intentional.
F-12	src/amm/Equestria1155.sol	INFO	reentrancy-events	onERC1155Received(...)	False positive; external receiver callback occurs at the correct stage of transfer flow.

4.2. Solhint Findings

Solhint Suppression Report

Rule	Affected Code / Scope	Justification
gas-strict-inequalities	if (amountOut >= reserveOut) revert InsufficientLiquidity();	Using strict '>' would allow draining one side of the pool when amountOut == reserveOut. Adding +/-1 or modifying constants only to satisfy the rule would be clunky and incorrect.
function-max-lines	Global configuration	The default limit of 50 lines is too restrictive for the project; increased to 80.
use-natspec	Global configuration	Full NatSpec refactoring would require excessive effort and large-scale annotation.
func-visibility	Global configuration	Deprecated warning; not applicable in newer Solidity versions.
import-path-check	Global configuration	Handled through foundry.toml configuration; considered a false positive.
immutable-vars-naming	Global configuration	Conflicts with Slither naming checks.

const-name-snakecase	Global configuration	Conflicts with Slither naming checks.
no-empty-blocks	Global configuration	Some overridden functions intentionally have empty bodies.
no-inline-assembly	Global configuration	Inline assembly is required by the assignment; warnings are expected and intentional.

5. Detailed Findings

F-01 — Reentrancy Risk During Random Request [MEDIUM]

Contract: Equestria1155.sol

Function: craft(uint256)

Line: 181-191

Description:

The contract performs an external call to the Chainlink VRF coordinator before updating internal state variables. Specifically, requestRandomWords() is called before pendingCraftUser and pendingCraftPonyId mappings are updated.

Although the current implementation is considered “benign reentrancy” by Slither and no direct exploit was identified, updating state after external interaction violates the Checks-Effects-Interactions pattern and may introduce future attack vectors if additional logic is added later.

Proof of Concept:

Slither detector:

Reentrancy in Equestria1155.craft(uint256)

External call:

requestRandomWords(...)

State updates after call:

pendingCraftUser[requestId] = user

pendingCraftPonyId[requestId] = ponyId

Recommendation:

Move all state updates before the external VRF request call or add explicit reentrancy protection using ReentrancyGuard.

Status: Open

F-02 — Missing Zero Address Validation [LOW]

Contract: Equestria1155.sol

Function: setGovernor(address)

Line: 141-143

Description:

The `setGovernor()` function does not validate whether the provided governor address equals `address(0)`. Accidentally setting the governor to the zero address would permanently disable privileged governance functionality.

Proof of Concept:

```
function setGovernor(address gameGovernor_) external onlyOwner {  
    gameGovernor = gameGovernor_;  
}
```

The function accepts `address(0)` without validation.

Recommendation:

Add a zero address check before assignment:

```
require(gameGovernor_ != address(0), "zero address");
```

Status: Open

6. Known Risk Areas

Area	Risk	Reason
ERC-1155 batch callbacks	Reentrancy	<code>onERC1155BatchReceived</code> can re-enter Equestria1155 or NFTRentalVault before state is finalised
Equestria1155.craft() burn/VRF	Atomicity	Tokens burned before VRF request; if VRF subscription runs out of LINK, tokens are consumed but no pony is ever minted
PriceFeedConsumer staleness	Oracle manipulation	Stale price window (10 min) could allow AMM pricing lag on mainnet; consider TWAP fallback
AMM constant-product invariant	Sandwich attacks	No MEV protection; users should set tight <code>minAmountOut</code> ; invariant fuzz tests validate $k \geq \text{initial}$
UUPS <code>_authorizeUpgrade</code>	Upgrade authority	GameTokenV1/V2 upgrade gated to owner; on testnet owner = deployer EOA; must transfer to Timelock in production
VRFConsumer <code>requestRandom()</code>	Request tracking bug	Potential <code>requestId</code> tracking mismatch in VRFConsumer
NFTRentalVault boost double-application	Share inflation	<code>_convertToShares</code> and <code>_deposit</code> both check <code>isRenting</code> — Potential double accounting in boost application logic; requires careful review

GovernanceToken distribution	Centralisation at deploy	On testnet all 1M tokens minted to deployer address; team must transfer to real vesting/treasury/airdrop contracts post-deploy
------------------------------	--------------------------	--

7. Recommendations Summary

Priority	ID	Recommendation
Medium	F-01, F-02	Consider using a fixed-point math library (e.g. PRBMath) for production deployments to eliminate division truncation risk. For this protocol the Uniswap V2 pattern is acceptable and well-understood.
Informational	F-03, F-07	Timestamp-based comparisons are unavoidable for vesting and oracle staleness checks. Miners can manipulate timestamps by ~15 seconds — acceptable given the time windows involved (days for vesting, hours for staleness).
Informational	F-04	Strict equality on zero amount is intentional and correct — no action required.
Informational	F-05	Low-level ETH call is the correct pattern per EIP best practices. <code>transfer()</code> and <code>send()</code> are explicitly avoided as required. Return value is checked.
Informational	F-06, F-08, F-11, F-12	Reentrancy findings are false positives. All state-changing functions use ReentrancyGuard or follow CEI pattern. No action required.
Informational	F-09	Yul assembly is intentional and required by project specification. Marked memory-safe. No action required.
Informational	F-10	<code>nonZeroAddress</code> modifier already enforces the zero-check. Slither cannot detect custom modifiers as equivalent to inline checks. No action required.

8.

Conclusion

The Equestria Protocol smart contract codebase was audited across all contracts in the ``src/`` directory. No Critical or High severity findings were identified. Two Medium findings were identified relating to integer division ordering in the AMM — both are intentional design decisions following the Uniswap V2 pattern and are mitigated by the protocol's L2 deployment context and test coverage.

All Informational findings have been reviewed and justified. The codebase demonstrates consistent application of security best practices including the Checks-Effects-Interactions pattern, OpenZeppelin ReentrancyGuard, SafeERC20 for all ERC-20 interactions, and role-based access control throughout.

The protocol is considered safe for testnet deployment. A follow-up audit is recommended before any mainnet deployment, particularly focusing on the AMM liquidity math and the async VRF crafting flow.

Overall protocol risk: Low

9. Automated Tools Output

9.1 Slither

slither . --exclude-dependencies --filter-paths "lib/"

slither . --exclude-dependencies --filter-paths "lib/"

'forge clean' running (wd: C:\Users\w2\Desktop\bt2-final\bt2-final)

'forge config --json' running

'forge build --build-info --skip ./test/** ./script/** --force' running (wd: C:\Users\w2\Desktop\bt2-final\bt2-final)

INFO:Detectors:

Detector: divide-before-multiply

AMM.addLiquidity(uint256,uint256,uint256,uint256) (src/amm/AMM.sol#56-110) performs a multiplication on the result of a division:

- amountBOptimal = (amountADesired * reserveB) / reserveA (src/amm/AMM.sol#75-76)
- amountB = amountBOptimal (src/amm/AMM.sol#86-87)
- liquidity = _min((amountA * supply) / reserveA,(amountB * supply) / reserveB) (src/amm/AMM.sol#97-99)

AMM.addLiquidity(uint256,uint256,uint256,uint256) (src/amm/AMM.sol#56-110) performs a multiplication on the result of a division:

- amountBOptimal = (amountADesired * reserveB) / reserveA (src/amm/AMM.sol#75-76)
- amountB = amountBOptimal (src/amm/AMM.sol#86-87)
- liquidity = _sqrt(amountA * amountB) (src/amm/AMM.sol#94-96)

Reference: <https://github.com/crytic/slither/wiki/Detector-Documentation#divide-before-multiply>

INFO:Detectors:

Detector: incorrect-equality

TokenVesting.release() (src/dao/TokenVesting.sol#42-56) uses a dangerous strict equality:

- amount == 0 (src/dao/TokenVesting.sol#53)

Reference: <https://github.com/crytic/slither/wiki/Detector-Documentation#dangerous-strict-equalities>

INFO:Detectors:

Detector: missing-zero-check

Equestria1155.setGovernor(address).gameGovernor_ (src/amm/Equestria1155.sol#141) lacks a zero-check on :

- gameGovernor = gameGovernor_ (src/amm/Equestria1155.sol#143)

Reference: <https://github.com/crytic/slither/wiki/Detector-Documentation#missing-zero-address-validation>

INFO:Detectors:

Detector: reentrancy-benign

Reentrancy in Equestria1155.craft(uint256) (src/amm/Equestria1155.sol#154-191):

External calls:

- requestId =

s_vrfCoordinator.requestRandomWords(VRFV2PlusClient.RandomWordsRequest({keyHash:keyHash,subId:subscriptionId,requestConfirmations:REQUEST_CONFIRMATIONS,callbackGasLimit:CALLBACK_GAS_LIMIT,numWords:1,extraArgs:VRFV2PlusClient._argsToBytes(VRFV2PlusClient.ExtraArgsV1({nativePayment:false})))) (src/amm/Equestria1155.sol#181-189)

State variables written after the call(s):

- pendingCraftPonyId[requestId] = ponyId (src/amm/Equestria1155.sol#190-191)
- pendingCraftUser[requestId] = user (src/amm/Equestria1155.sol#189-190)

Reentrancy in VRFConsumer.requestRandom() (src/chainlink/VRFConsumer.sol#22-44):

External calls:

- requestId =

s_vrfCoordinator.requestRandomWords(VRFV2PlusClient.RandomWordsRequest({keyHash:keyHash,subId:subscriptionId,requestConfirmations:REQUEST_CONFIRMATIONS,callbackGasLimit:CALLBACK_GAS_LIMIT,numWords:1,extraArgs:VRFV2PlusClient._argsToBytes(VRFV2PlusClient.ExtraArgsV1({nativePayment:false})))) (src/chainlink/VRFConsumer.sol#22-44)

ords:NUM_WORDS,extraArgs:VRFV2PlusClient._argsToBytes(VRFV2PlusClient.ExtraArgsV1({nativePayment:false})))) (src/chainlink/VRFConsumer.sol#32-40)

State variables written after the call(s):

- requestToSender[requestId] = msg.sender (src/chainlink/VRFConsumer.sol#40-43)

Reference: <https://github.com/crytic/slither/wiki/Detector-Documentation#reentrancy-vulnerabilities-3>

INFO:Detectors:

Detector: reentrancy-events

Reentrancy in Equestria1155.fulfillRandomWords(uint256,uint256[]) (src/amm/Equestria1155.sol#191-208):

External calls:

- _mint(user,ponyId,amount,) (src/amm/Equestria1155.sol#205-206)
- response = IERC1155Receiver(to).onERC1155Received(operator,from,id,value,data)

(src/amm/Equestria1155.sol#362-363)

Event emitted after the call(s):

- CraftFulfilled(requestId,user,ponyId) (src/amm/Equestria1155.sol#206-208)

Reentrancy in VRFConsumer.requestRandom() (src/chainlink/VRFConsumer.sol#22-44):

External calls:

- requestId =

s_vrfCoordinator.requestRandomWords(VRFV2PlusClient.RandomWordsRequest({keyHash:keyHash,subId:subscriptionId,requestConfirmations:REQUEST_CONFIRMATIONS,callbackGasLimit:CALLBACK_GAS_LIMIT,numWords:NUM_WORDS,extraArgs:VRFV2PlusClient._argsToBytes(VRFV2PlusClient.ExtraArgsV1({nativePayment:false})))) (src/chainlink/VRFConsumer.sol#32-40)

Event emitted after the call(s):

- RandomRequested(requestId,msg.sender) (src/chainlink/VRFConsumer.sol#43-44)

Reference: <https://github.com/crytic/slither/wiki/Detector-Documentation#reentrancy-vulnerabilities-4>

INFO:Detectors:

Detector: timestamp

PriceFeedConsumer.getLatestPrice() (src/chainlink/PriceFeedConsumer.sol#16-32) uses timestamp for comparisons

Dangerous comparisons:

- block.timestamp - updatedAt_ > stalenessThreshold (src/chainlink/PriceFeedConsumer.sol#28-29)

TokenVesting.vestedAmount() (src/dao/TokenVesting.sol#25-42) uses timestamp for comparisons

Dangerous comparisons:

- block.timestamp < start (src/dao/TokenVesting.sol#34-35)
- block.timestamp >= start + DURATION (src/dao/TokenVesting.sol#39-40)

TokenVesting.release() (src/dao/TokenVesting.sol#42-56) uses timestamp for comparisons

Dangerous comparisons:

- amount == 0 (src/dao/TokenVesting.sol#53)

Reference: <https://github.com/crytic/slither/wiki/Detector-Documentation#block-timestamp>

INFO:Detectors:

Detector: assembly

AMM.getAmountOutYul(uint256,uint256,uint256) (src/amm/AMM.sol#192-210) uses assembly

- INLINE ASM (src/amm/AMM.sol#201-210)

Reference: <https://github.com/crytic/slither/wiki/Detector-Documentation#assembly-usage>

INFO:Detectors:

Detector: low-level-calls

Low level call in Treasury.withdrawETH(address,uint256) (src/dao/Treasury.sol#32-41):

- (success,None) = to.call{value: amount}() (src/dao/Treasury.sol#39-40)

Reference: <https://github.com/cryptic/slither/wiki/Detector-Documentation#low-level-calls>

INFO:Slither:. analyzed (87 contracts with 101 detectors), 13 result(s) found

9.2 Foundry Coverage

forge coverage --report lcov

Compiling 127 files with Solc 0.8.33

Solc 0.8.33 finished in 12.14s

Compiler run successful with warnings:

Warning (5574): Contract code size is 30599 bytes and exceeds 24576 bytes (a limit introduced in Spurious Dragon). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

--> src/dao/MyGovernor.sol:17:1:

```
|  
17 | contract MyGovernor is  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (5574): Contract code size is 184346 bytes and exceeds 24576 bytes (a limit introduced in Spurious Dragon). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

--> script/Deploy.s.sol:31:1:

```
|  
31 | contract Deploy is Script {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 184613 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

--> script/Deploy.s.sol:31:1:

```
|  
31 | contract Deploy is Script {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 72622 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

--> test/amm/AMM.t.sol:57:1:

```
|  
57 | contract AMMTest is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 53995 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

--> test/amm/Equestria1155.t.sol:60:1:

```
|  
60 | contract Equestria1155Test is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 85794 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

```
--> test/dao/BoxTest.t.sol:12:1:
```

```
|  
12 | contract BoxTest is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 104088 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

```
--> test/dao/FullFlow.t.sol:14:1:
```

```
|  
14 | contract FullFlowTest is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 49603 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

```
--> test/dao/GovernanceToken.t.sol:8:1:
```

```
|  
8 | contract GovernanceTokenTest is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 98736 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

```
--> test/dao/MyGovernor.t.sol:13:1:
```

```
|  
13 | contract MyGovernorTest is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 50889 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

```
--> test/factory/GameTokenFactory.t.sol:12:1:
```

```
|  
12 | contract GameTokenFactoryTest is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 108019 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

--> test/invariant/Invariants.t.sol:77:1:

```
|  
77 | contract InvariantTests is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Warning (3860): Contract initcode size is 89024 bytes and exceeds 49152 bytes (a limit introduced in Shanghai). This contract may not be deployable on Mainnet. Consider enabling the optimizer (with a low "runs" value!), turning off revert strings, or using libraries.

--> test/vault/NFTRentalVault.t.sol:12:1:

```
|  
12 | contract NFTRentalVaultTest is Test {  
    | ^ (Relevant source part starts here and spans across multiple lines).
```

Analysing contracts...

Running tests...

Ran 4 tests for test/vulnerability/VulnerabilityTest.t.sol:VulnerabilityTest
[PASS] test_accessControl_fixed_onlyOwnerCanWithdraw() (gas: 314441)
[PASS] test_accessControl_vulnerable_anyoneCanWithdraw() (gas: 281845)
[PASS] test_reentrancy_fixed_attackReverts() (gas: 703942)
[PASS] test_reentrancy_vulnerable_attackSucceeds() (gas: 831976)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 1.91ms (1.58ms CPU time)

Ran 5 tests for test/dao/Treasury.t.sol:TreasuryTest
[PASS] test_receiveETH() (gas: 17236)
[PASS] test_withdrawETH_onlyTimelock() (gas: 11577)
[PASS] test_withdrawETH_works() (gas: 49445)
[PASS] test_withdrawToken_onlyTimelock() (gas: 52596)
[PASS] test_withdrawToken_works() (gas: 68512)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 3.60ms (2.50ms CPU time)

Ran 4 tests for test/dao/TokenVesting.t.sol:TokenVestingTest
[PASS] testCannotReleaseBeforeVesting() (gas: 20067)
[PASS] testFullVestingAndRelease() (gas: 87753)
[PASS] testInitialVesting() (gas: 18109)
[PASS] testLinearVestingMidpoint() (gas: 21466)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 6.81ms (5.37ms CPU time)

Ran 1 test for test/dao/FullFlow.t.sol:FullFlowTest
[PASS] testFullDAOFlow() (gas: 804065)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 18.66ms (9.91ms CPU time)

Ran 7 tests for test/chainlink/VRFConsumer.t.sol:VRFConsumerTest
[PASS] testFuzz_fulfillRandomWords_anyWord(uint256) (runs: 256, μ : 148054, $\sim$: 148676)

[PASS] test_fulfillRandomWords_emitsEvent() (gas: 148405)
[PASS] test_fulfillRandomWords_storesResult() (gas: 148834)
[PASS] test_multipleRequests_independentResults() (gas: 236802)
[PASS] test_requestRandom_emitsEvent() (gas: 115858)
[PASS] test_requestRandom_returnsRequestId() (gas: 113059)
[PASS] test_requestRandom_storesSender() (gas: 115793)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 312.68ms (68.83ms CPU time)

Ran 12 tests for test/dao/MyGovernor.t.sol:MyGovernorTest

[PASS] test_defeated() (gas: 115137)
[PASS] test_defeated_quorum_not_met() (gas: 179002)
[PASS] test_delegation() (gas: 86575)
[PASS] test_execute() (gas: 410722)
[PASS] test_proposal_threshold_blocks_low_balance() (gas: 167871)
[PASS] test_propose() (gas: 86183)
[PASS] test_queue() (gas: 345324)
[PASS] test_state_active() (gas: 97045)
[PASS] test_succeeded_state() (gas: 234042)
[PASS] test_vote_abstain() (gas: 162778)
[PASS] test_vote_against() (gas: 162450)
[PASS] test_vote_for() (gas: 162658)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 312.78ms (34.04ms CPU time)

Ran 3 tests for test/fork/Fork.t.sol:ForkTest

[PASS] test_fork_chainlinkPriceFeed() (gas: 371016)
[PASS] test_fork_chainlinkStalenessCheck() (gas: 374667)
[PASS] test_fork_usdcTransfer() (gas: 54848)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 2.27s (4.75s CPU time)

Ran 11 tests for test/chainlink/PriceFeedConsumer.t.sol:PriceFeedConsumerTest

[PASS] test_getLatestPrice_exactThresholdNotStale() (gas: 20199)
[PASS] test_getLatestPrice_returnsPrice() (gas: 16398)
[PASS] test_getLatestPrice_returnsUpdatedAt() (gas: 16604)
[PASS] test_getLatestPrice_revertsOnNegativePrice() (gas: 27115)
[PASS] test_getLatestPrice_revertsOnZeroPrice() (gas: 22059)
[PASS] test_getLatestPrice_revertsWhenStale() (gas: 20601)
[PASS] test_mockAggregator_description() (gas: 7913)
[PASS] test_mockAggregator_getRoundData() (gas: 12187)
[PASS] test_mockAggregator_setPrice() (gas: 24029)
[PASS] test_mockAggregator_setUpdatedAt_makesStale() (gas: 27617)
[PASS] test_mockAggregator_version() (gas: 6837)
Suite result: ok. 11 passed; 0 failed; 0 skipped; finished in 82.00s (3.77ms CPU time)

Ran 20 tests for test/factory/GameTokenFactory.t.sol:GameTokenFactoryTest

[PASS] test_create2_deploysToken() (gas: 997432)
[PASS] test_create2_deterministicAddress() (gas: 1006056)
[PASS] test_create2_emitsEvent() (gas: 1003487)
[PASS] test_create2_revertsOnSaltReuse() (gas: 1040461293)
[PASS] test_create_deploysToken() (gas: 996015)
[PASS] test_create_differentAddressEachTime() (gas: 1964826)
[PASS] test_create_emitsEvent() (gas: 1001725)
[PASS] test_create_ownerIsCallerMessage() (gas: 998142)
[PASS] test_create_trackedInAllTokens() (gas: 1965871)
[PASS] test_proxy_cannotReinitialize() (gas: 232584)
[PASS] test_proxy_initialize() (gas: 239678)
[PASS] test_proxy_mint() (gas: 282648)
[PASS] test_proxy_mintOnlyOwner() (gas: 230260)
[PASS] test_proxy_version_v1() (gas: 231653)
[PASS] test_upgrade_onlyOwner() (gas: 234206)
[PASS] test_upgrade_statePreserved() (gas: 300149)
[PASS] test_upgrade_toV2() (gas: 247058)
[PASS] test_upgrade_v2_capEnforced() (gas: 276326)
[PASS] test_upgrade_v2_initializeCap() (gas: 271329)
[PASS] test_upgrade_v2_mintUnderCap() (gas: 325097)
Suite result: ok. 20 passed; 0 failed; 0 skipped; finished in 82.00s (29.87ms CPU time)

Ran 18 tests for test/vault/NFTRentalVault.t.sol:NFTRentalVaultTest

[PASS] testFuzz_depositRedeem_roundingInvariant(uint256) (runs: 256, μ : 115420, $\sim$: 115645)
[PASS] testFuzz_injectYield_increasesTotalAssets(uint256) (runs: 256, μ : 159958, $\sim$: 160068)
[PASS] test_boost_givesMoreShares() (gas: 332576)
[PASS] test_deposit_assetsTransferred() (gas: 128040)
[PASS] test_deposit_basic() (gas: 122399)
[PASS] test_injectYield_increasesAssets() (gas: 158145)
[PASS] test_injectYield_onlyOwner() (gas: 13572)
[PASS] test_previewDeposit_matchesActual() (gas: 128921)
[PASS] test_redeem_basic() (gas: 118582)
[PASS] test_setBoostBps_belowBaseReverts() (gas: 16280)
[PASS] test_setBoostBps_onlyOwner() (gas: 13066)
[PASS] test_setBoostBps_updated() (gas: 24469)
[PASS] test_stakeNFT() (gas: 161709)
[PASS] test_stakeNFT_doubleStakeReverts() (gas: 165941)
[PASS] test_stakeNFT_wrongId() (gas: 52518)
[PASS] test_unstakeNFT() (gas: 151286)
[PASS] test_unstakeNFT_notRenterReverts() (gas: 169273)
[PASS] test_withdraw_basic() (gas: 122311)
Suite result: ok. 18 passed; 0 failed; 0 skipped; finished in 82.00s (672.85ms CPU time)

Ran 13 tests for test/amm/Equestria1155.t.sol:Equestria1155Test

```

[PASS] testBalanceOfBatch() (gas: 33662)
[PASS] testCraftDoubleDropOnLuckyRoll() (gas: 324904)
[PASS] testCraftRevertsOnInsufficientResources() (gas: 78685)
[PASS] testCraftSuccessMintsNFTAndBurnsResources() (gas: 359080)
[PASS] testMintBatchAndBalances() (gas: 72020)
[PASS] testMintSingleFungible() (gas: 32828)
[PASS] testSafeBatchTransferFromApprovedOperatorToGoodReceiver() (gas: 153376)
[PASS] testSafeTransferFromApprovedOperatorToEOA() (gas: 77770)
[PASS] testSetGovernor_onlyOwner() (gas: 16039)
[PASS] testSetRecipe_governorCanUpdate() (gas: 96815)
[PASS] testSetRecipe_onlyGovernor() (gas: 18516)
[PASS] testTransferToRejectingContractReverts() (gas: 98707)
[PASS] testUriSubstitution() (gas: 195016)
Suite result: ok. 13 passed; 0 failed; 0 skipped; finished in 82.00s (18.09ms CPU time)

```

Ran 44 tests for test/amm/AMM.t.sol:AMMTest

```

[PASS] invariant_kNeverDecreases() (runs: 256, calls: 128000, reverts: 989)

```

```

┌-----+-----+-----+-----┐
| Contract | Selector | Calls | Reverts | Discards |
+=====+
| AMMHandler | addLiquidity | 42773 | 328 | 0 |
└-----+-----+-----+-----┘
| AMMHandler | removeLiquidity | 42617 | 661 | 0 |
└-----+-----+-----+-----┘
| AMMHandler | swap | 42610 | 0 | 0 |
┌-----+-----+-----+-----┐

```

```

[PASS] testFuzz_addRemoveLiquidity_noLoss(uint256) (runs: 256, μ: 260498, ~: 261368)
[PASS] testFuzz_getAmountOut_alwaysLessThanInput(uint256) (runs: 256, μ: 9904, ~: 9351)
[PASS] testFuzz_getAmountOut_yulMatchesSolidity(uint256,uint256,uint256) (runs: 256, μ: 13357, ~: 13158)
[PASS] testFuzz_removeLiquidity_neverExceedsReserves(uint256) (runs: 256, μ: 248685, ~: 249554)
[PASS] testFuzz_swap_constantProduct(uint256) (runs: 256, μ: 299915, ~: 300049)
[PASS] test_addLiquidity_emitsEvent() (gas: 243489)
[PASS] test_addLiquidity_initial() (gas: 243271)
[PASS] test_addLiquidity_mintsLPTokens() (gas: 244602)
[PASS] test_addLiquidity_revertsZeroAmount() (gas: 19996)
[PASS] test_addLiquidity_slippageA() (gas: 252130)
[PASS] test_addLiquidity_slippageB() (gas: 252731)
[PASS] test_addLiquidity_subsequentKeepsRatio() (gas: 326805)
[PASS] test_benchmark_getAmountOut_solidity() (gas: 8552)
[PASS] test_benchmark_getAmountOut_yul() (gas: 7434)
[PASS] test_constructor_deploysLPToken() (gas: 3493)
[PASS] test_constructor_revertsIdenticalTokens() (gas: 67947)

```

[PASS] test_constructor_revertsZeroAddress() (gas: 67188)
 [PASS] test_constructor_setsTokens() (gas: 13728)
 [PASS] test_getAmountOut_basic() (gas: 8739)
 [PASS] test_getAmountOut_matchesYul() (gas: 10797)
 [PASS] test_getAmountOut_revertsZeroReserve() (gas: 11362)
 [PASS] test_lpToken_approve() (gas: 44549)
 [PASS] test_lpToken_decimals() (gas: 7376)
 [PASS] test_lpToken_transfer() (gas: 280937)
 [PASS] test_lpToken_transferFrom() (gas: 269401)
 [PASS] test_lpToken_transferFrom_maxAllowanceNotReduced() (gas: 287161)
 [PASS] test_lpToken_transferFrom_reducesAllowance() (gas: 312577)
 [PASS] test_lpToken_transferFrom_revertsInsufficientAllowance() (gas: 250233)
 [PASS] test_lpToken_transfer_revertsInsufficientBalance() (gas: 18691)
 [PASS] test_quotes_nonZeroAfterLiquidity() (gas: 247730)
 [PASS] test_quotes_zeroBeforeLiquidity() (gas: 12425)
 [PASS] test_removeLiquidity_basic() (gas: 248132)
 [PASS] test_removeLiquidity_emitsEvent() (gas: 248964)
 [PASS] test_removeLiquidity_reservesDecreased() (gas: 248908)
 [PASS] test_removeLiquidity_revertsZero() (gas: 248078)
 [PASS] test_removeLiquidity_slippage() (gas: 279932)
 [PASS] test_swap_AtoB() (gas: 298057)
 [PASS] test_swap_BtoA() (gas: 298392)
 [PASS] test_swap_emitsEvent() (gas: 294875)
 [PASS] test_swap_revertsInvalidToken() (gas: 250060)
 [PASS] test_swap_revertsSlippage() (gas: 254314)
 [PASS] test_swap_revertsZeroAmount() (gas: 252675)
 [PASS] test_swap_updatesReserves() (gas: 291527)
 Suite result: ok. 44 passed; 0 failed; 0 skipped; finished in 82.00s (59.69s CPU time)

Ran 7 tests for test/dao/GovernanceToken.t.sol:GovernanceTokenTest

[PASS] testDelegation() (gas: 97880)
 [PASS] testDelegationUpdate() (gas: 189772)
 [PASS] testFuzz_delegate_votingPowerMatchesBalance(uint256) (runs: 256, μ : 126755, $\sim$: 126949)
 [PASS] testFuzz_transfer_conservesSupply(uint256) (runs: 256, μ : 45318, $\sim$: 44977)
 [PASS] testInitialDistribution() (gas: 32637)
 [PASS] testPermit() (gas: 329804)
 [PASS] testVotingPowerSnapshot() (gas: 98710)
 Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 82.06s (82.18s CPU time)

Ran 9 tests for test/invariant/Invariants.t.sol:InvariantTests

[PASS] invariant_amm_kNeverDecreases() (runs: 256, calls: 128000, reverts: 473)

(-----+-----+-----+-----+-----)
 | Contract | Selector | Calls | Reverts | Discards |

+=====+					
AMMHandler	addLiquidity	18299	121	0	
-----+-----+-----+-----+-----					
AMMHandler	removeLiquidity	18323	352	0	
-----+-----+-----+-----+-----					
AMMHandler	swap	18441	0	0	
-----+-----+-----+-----+-----					
TreasuryHandler	deposit	18333	0	0	
-----+-----+-----+-----+-----					
TreasuryHandler	withdraw	18189	0	0	
-----+-----+-----+-----+-----					
VaultHandler	deposit	18167	0	0	
-----+-----+-----+-----+-----					
VaultHandler	redeem	18248	0	0	
-----+-----+-----+-----+-----					

[PASS] invariant_amm_lpSupplyNonZeroWhenReservesNonZero() (runs: 256, calls: 128000, reverts: 464)

-----+-----+-----+-----+-----					
Contract	Selector	Calls	Reverts	Discards	
+=====+					
AMMHandler	addLiquidity	18299	109	0	
-----+-----+-----+-----+-----					
AMMHandler	removeLiquidity	18323	355	0	
-----+-----+-----+-----+-----					
AMMHandler	swap	18441	0	0	
-----+-----+-----+-----+-----					
TreasuryHandler	deposit	18333	0	0	
-----+-----+-----+-----+-----					
TreasuryHandler	withdraw	18189	0	0	
-----+-----+-----+-----+-----					
VaultHandler	deposit	18167	0	0	
-----+-----+-----+-----+-----					
VaultHandler	redeem	18248	0	0	
-----+-----+-----+-----+-----					

[PASS] invariant_govToken_totalSupplyConstant() (runs: 256, calls: 128000, reverts: 478)

-----+-----+-----+-----+-----					
Contract	Selector	Calls	Reverts	Discards	
+=====+					
AMMHandler	addLiquidity	18299	122	0	
-----+-----+-----+-----+-----					
AMMHandler	removeLiquidity	18323	356	0	

+-----+-----+-----+-----+-----+				
Contract	Selector	Calls	Reverts	Discards
+=====+				
AMMHandler	swap	18441	0	0
+-----+-----+-----+-----+-----+				
TreasuryHandler	deposit	18333	0	0
+-----+-----+-----+-----+-----+				
TreasuryHandler	withdraw	18189	0	0
+-----+-----+-----+-----+-----+				
VaultHandler	deposit	18167	0	0
+-----+-----+-----+-----+-----+				
VaultHandler	redeem	18248	0	0
+-----+-----+-----+-----+-----+				

[PASS] invariant_treasury_balanceEqualsDepositedMinusWithdrawn() (runs: 256, calls: 128000, reverts: 471)

+-----+-----+-----+-----+-----+				
Contract	Selector	Calls	Reverts	Discards
+=====+				
AMMHandler	addLiquidity	18299	115	0
+-----+-----+-----+-----+-----+				
AMMHandler	removeLiquidity	18323	356	0
+-----+-----+-----+-----+-----+				
AMMHandler	swap	18441	0	0
+-----+-----+-----+-----+-----+				
TreasuryHandler	deposit	18333	0	0
+-----+-----+-----+-----+-----+				
TreasuryHandler	withdraw	18189	0	0
+-----+-----+-----+-----+-----+				
VaultHandler	deposit	18167	0	0
+-----+-----+-----+-----+-----+				
VaultHandler	redeem	18248	0	0
+-----+-----+-----+-----+-----+				

[PASS] invariant_treasury_balanceNonNegative() (runs: 256, calls: 128000, reverts: 493)

+-----+-----+-----+-----+-----+				
Contract	Selector	Calls	Reverts	Discards
+=====+				
AMMHandler	addLiquidity	18299	128	0
+-----+-----+-----+-----+-----+				
AMMHandler	removeLiquidity	18323	365	0
+-----+-----+-----+-----+-----+				
AMMHandler	swap	18441	0	0
+-----+-----+-----+-----+-----+				
TreasuryHandler	deposit	18333	0	0

-----+-----+-----+-----+-----				
TreasuryHandler	withdraw	18189	0	0
-----+-----+-----+-----+-----				
VaultHandler	deposit	18167	0	0
-----+-----+-----+-----+-----				
VaultHandler	redeem	18248	0	0
-----+-----+-----+-----+-----				

[PASS] invariant_vault_redeemNeverExceedsDeposit() (runs: 256, calls: 128000, reverts: 465)

-----+-----+-----+-----+-----				
Contract	Selector	Calls	Reverts	Discards
+=====+				
AMMHandler	addLiquidity	18299	131	0
-----+-----+-----+-----+-----				
AMMHandler	removeLiquidity	18323	334	0
-----+-----+-----+-----+-----				
AMMHandler	swap	18441	0	0
-----+-----+-----+-----+-----				
TreasuryHandler	deposit	18333	0	0
-----+-----+-----+-----+-----				
TreasuryHandler	withdraw	18189	0	0
-----+-----+-----+-----+-----				
VaultHandler	deposit	18167	0	0
-----+-----+-----+-----+-----				
VaultHandler	redeem	18248	0	0
-----+-----+-----+-----+-----				

[PASS] invariant_vault_totalAssetsMatchesBalance() (runs: 256, calls: 128000, reverts: 504)

-----+-----+-----+-----+-----				
Contract	Selector	Calls	Reverts	Discards
+=====+				
AMMHandler	addLiquidity	18299	137	0
-----+-----+-----+-----+-----				
AMMHandler	removeLiquidity	18323	367	0
-----+-----+-----+-----+-----				
AMMHandler	swap	18441	0	0
-----+-----+-----+-----+-----				
TreasuryHandler	deposit	18333	0	0
-----+-----+-----+-----+-----				
TreasuryHandler	withdraw	18189	0	0
-----+-----+-----+-----+-----				
VaultHandler	deposit	18167	0	0

Contract	Selector	Calls	Reverts	Discards
VaultHandler	redeem	18248	0	0

[PASS] invariant_vault_totalAssetsNonNegative() (runs: 256, calls: 128000, reverts: 475)

Contract	Selector	Calls	Reverts	Discards
AMMHandler	addLiquidity	18299	116	0
AMMHandler	removeLiquidity	18323	359	0
AMMHandler	swap	18441	0	0
TreasuryHandler	deposit	18333	0	0
TreasuryHandler	withdraw	18189	0	0
VaultHandler	deposit	18167	0	0
VaultHandler	redeem	18248	0	0

[PASS] invariant_vault_totalSupplyNonNegative() (runs: 256, calls: 128000, reverts: 493)

Contract	Selector	Calls	Reverts	Discards
AMMHandler	addLiquidity	18299	123	0
AMMHandler	removeLiquidity	18323	370	0
AMMHandler	swap	18441	0	0
TreasuryHandler	deposit	18333	0	0
TreasuryHandler	withdraw	18189	0	0
VaultHandler	deposit	18167	0	0
VaultHandler	redeem	18248	0	0

Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 93.73s (578.73s CPU time)

Ran 14 test suites in 93.73s (588.73s CPU time): 158 tests passed, 0 failed, 0 skipped (158 total tests)

Wrote LCOV report.

9.3 Gas Report

forge test --gas-report

Ran 4 tests for test/dao/TokenVesting.t.sol:TokenVestingTest

[PASS] testCannotReleaseBeforeVesting() (gas: 37766)

[PASS] testFullVestingAndRelease() (gas: 109152)

[PASS] testInitialVesting() (gas: 18813)

[PASS] testLinearVestingMidpoint() (gas: 16775)

Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 3.62ms (1.06ms CPU time)

Ran 4 tests for test/vulnerability/VulnerabilityTest.t.sol:VulnerabilityTest

[PASS] test_accessControl_fixed_onlyOwnerCanWithdraw() (gas: 269233)

[PASS] test_accessControl_vulnerable_anyoneCanWithdraw() (gas: 204636)

[PASS] test_reentrancy_fixed_attackReverts() (gas: 541023)

[PASS] test_reentrancy_vulnerable_attackSucceeds() (gas: 595168)

Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 1.93ms (2.28ms CPU time)

Ran 1 test for test/dao/FullFlow.t.sol:FullFlowTest

[PASS] testFullDAOFlow() (gas: 1070215)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 15.76ms (6.78ms CPU time)

Ran 12 tests for test/dao/MyGovernor.t.sol:MyGovernorTest

[PASS] test_defeated() (gas: 119406)

[PASS] test_defeated_quorum_not_met() (gas: 204103)

[PASS] test_delegation() (gas: 120558)

[PASS] test_execute() (gas: 507718)

[PASS] test_proposal_threshold_blocks_low_balance() (gas: 218434)

[PASS] test_propose() (gas: 91514)

[PASS] test_queue() (gas: 397578)

[PASS] test_state_active() (gas: 102269)

[PASS] test_succeeded_state() (gas: 278185)

[PASS] test_vote_abstain() (gas: 186545)

[PASS] test_vote_against() (gas: 186548)

[PASS] test_vote_for() (gas: 186536)

Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 14.85ms (17.74ms CPU time)

Ran 7 tests for test/chainlink/VRFConsumer.t.sol:VRFConsumerTest

[PASS] testFuzz_fulfillRandomWords_anyWord(uint256) (runs: 256, μ : 189779, $\sim$: 189880)

[PASS] test_fulfillRandomWords_emitsEvent() (gas: 188547)

[PASS] test_fulfillRandomWords_storesResult() (gas: 190001)

[PASS] test_multipleRequests_independentResults() (gas: 336238)

[PASS] test_requestRandom_emitsEvent() (gas: 130192)

[PASS] test_requestRandom_returnsRequestId() (gas: 127932)

[PASS] test_requestRandom_storesSender() (gas: 133410)

Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 215.28ms (216.30ms CPU time)

Ran 13 tests for test/amm/Equestria1155.t.sol:Equestria1155Test

[PASS] testBalanceOfBatch() (gas: 29130)

[PASS] testCraftDoubleDropOnLuckyRoll() (gas: 365631)

[PASS] testCraftRevertsOnInsufficientResources() (gas: 118587)

[PASS] testCraftSuccessMintsNFTAndBurnsResources() (gas: 431875)

[PASS] testMintBatchAndBalances() (gas: 99103)
[PASS] testMintSingleFungible() (gas: 57556)
[PASS] testSafeBatchTransferFromApprovedOperatorToGoodReceiver() (gas: 204996)
[PASS] testSafeTransferFromApprovedOperatorToEOA() (gas: 127397)
[PASS] testSetGovernor_onlyOwner() (gas: 35213)
[PASS] testSetRecipe_governorCanUpdate() (gas: 148358)
[PASS] testSetRecipe_onlyGovernor() (gas: 41459)
[PASS] testTransferToRejectingContractReverts() (gas: 141886)
[PASS] testUriSubstitution() (gas: 64508)
Suite result: ok. 13 passed; 0 failed; 0 skipped; finished in 229.01ms (10.56ms CPU time)

Ran 18 tests for test/vault/NFTRentalVault.t.sol:NFTRentalVaultTest
[PASS] testFuzz_depositRedeem_roundingInvariant(uint256) (runs: 256, μ : 162367, $\sim$: 162431)
[PASS] testFuzz_injectYield_increasesTotalAssets(uint256) (runs: 256, μ : 196931, $\sim$: 196943)
[PASS] test_boost_givesMoreShares() (gas: 385275)
[PASS] test_deposit_assetsTransferred() (gas: 142659)
[PASS] test_deposit_basic() (gas: 128351)
[PASS] test_injectYield_increasesAssets() (gas: 196534)
[PASS] test_injectYield_onlyOwner() (gas: 32942)
[PASS] test_previewDeposit_matchesActual() (gas: 141367)
[PASS] test_redeem_basic() (gas: 167782)
[PASS] test_setBoostBps_belowBaseReverts() (gas: 35191)
[PASS] test_setBoostBps_onlyOwner() (gas: 32396)
[PASS] test_setBoostBps_updated() (gas: 44178)
[PASS] test_stakeNFT() (gas: 187376)
[PASS] test_stakeNFT_doubleStakeReverts() (gas: 214246)
[PASS] test_stakeNFT_wrongId() (gas: 90316)
[PASS] test_unstakeNFT() (gas: 245520)
[PASS] test_unstakeNFT_notRenterReverts() (gas: 217236)
[PASS] test_withdraw_basic() (gas: 169792)
Suite result: ok. 18 passed; 0 failed; 0 skipped; finished in 254.19ms (482.25ms CPU time)

Ran 3 tests for test/fork/Fork.t.sol:ForkTest
[PASS] test_fork_chainlinkPriceFeed() (gas: 252872)
[PASS] test_fork_chainlinkStalenessCheck() (gas: 256285)
[PASS] test_fork_usdcTransfer() (gas: 103907)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 2.30s (4.80s CPU time)